

[6] Power - Generation

The schematic diagram illustrates the power generation section, featuring four voltage regulators connected to a +VBATD input. The regulators are:

- 5V ± 2.5% 600mA Buck Regulator (2.2MHz):** Utilizes the LMR36506M5C5RPERQ1 (U2). It includes a feedback network with FB2 (1k @ 100MHz) and a compensation network with C24 (1u X7R 10V) and R4 (0R). The output is +5V.
- 12V ± 1.5% 600mA Buck Regulator (2.1MHz):** Utilizes the LMR36006FSCQ (U3). It includes a feedback network with FB3 (1k @ 100MHz) and a compensation network with C26 (1u X7R 10V). The output is +12V.
- 3.3V ± 3% 500mA Buck Regulator (2.25MHz):** Utilizes the TPS62172DSG (U4). It includes a feedback network with FB4 (1k @ 100MHz) and a compensation network with C45 (2u2 X7R 100V), C52 (2u2 X7R 100V), and C56 (10u X7R 25V). The output is +3V3.
- 3.3V ± 0.5% 250mA Linear Regulator:** Utilizes the LP2992IM5-3.3/NOPB (U5). It includes a feedback network with C60 (1u X7R 25V) and C76 (10n X7R 100V). The output is +A3V3.

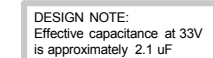
Design Notes:

- DESIGN NOTE: $L = (VIN - VOUT) / (f_{SW} \cdot K \cdot I_{OUT}) \cdot VOUT / VIN$. Computed with $VIN = 50V$, $K = 0.15$.
- DESIGN NOTE: Pi filter with inductor would give better EMC performance but has higher bulk capacitance requirements. Ferrite is a good compromise.
- DESIGN NOTE: MODE = LOW → Auto Mode.

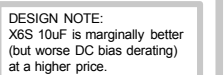
Layout Notes:

- LAYOUT NOTE: Route feedback traces away from switching node.
- LAYOUT NOTE: Minimize switching node loop areas.

Voltage Regulators



Test Points



Comments:	Company:	Variant:		Git Hash:
+12V (600 mA): Fan power +5V (600 mA): Gate Driver, LEDs +3V3 (500 mA): digital 3.3V +3V3A (250 mA): analog 3.3V	EPFL Xplore Research			5b68472
Board Name:		Project Name:		
Amulet Motion Controller		Chienpanzé		
Sheet Title:	File Name:	Designer:	Date:	Revision:
Power - Generation	Power - Generation.kicad_sch	Vincent Nguyen	2024-04-13	1.1.1+ (Unreleased)
Sheet Path:		Reviewer:	Size:	Sheet:
/Project Architecture/Power - Generation/			A3	6 of 21